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SUBSTRATE PROCESSING APPARATUS

Abstract

A substrate processing apparatus includes a reaction tube, a vacuum pipe integrally formed with the reaction tube, a housing configured to accommodate the reaction tube and the vacuum pipe therein, an exhaust pipe provided below the housing, and a fixing member configured to fix the exhaust pipe to the housing. The housing has a bottom portion configured to support the reaction tube, the bottom portion has an opening with an opening diameter larger than an outer diameter of the vacuum pipe, the vacuum pipe has a lower end inserted through the opening, and the exhaust pipe is fixed to the housing in a state in which an interior thereof communicates with an interior of the vacuum pipe.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-025088, filed on Feb. 22, 2024, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to a substrate processing apparatus.

BACKGROUND

[0003] In a batch-type vacuum processing apparatus, a technique is known in which a bellows body that is stretchable and bendable is provided at a connection portion between an exhaust port of a reaction tube and a vacuum pipe (see, for example, Patent Document 1).

PRIOR ART DOCUMENT

Patent Document

[0004] Patent Document 1: Japanese Patent Laid-Open Publication No. 2012-104755

SUMMARY

[0005] According to one embodiment of the present disclosure, a substrate processing apparatus includes a reaction tube, a vacuum pipe integrally formed with the reaction tube, a housing configured to accommodate the reaction tube and the vacuum pipe therein, an exhaust pipe provided below the housing, and a fixing member configured to fix the exhaust pipe to the housing. The housing has a bottom portion configured to support the reaction tube, the bottom portion has an opening with an opening diameter larger than an outer diameter of the vacuum pipe, the vacuum pipe has a lower end inserted through the opening, and the exhaust pipe is fixed to the housing in a state in which an interior thereof communicates with an interior of the vacuum pipe.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0006] The accompanying drawings, which are incorporated in and constitute a part of the specification, illustrate embodiments of the present disclosure, and together with the general description given above and the detailed description of the embodiments given below, serve to explain the principles of the present disclosure.

[0007] FIG. 1 is a perspective view showing a substrate processing apparatus according to an embodiment.

[0008] FIG. 2 is a perspective view showing the substrate processing apparatus according to the embodiment.

[0009] FIG. 3 is a vertical cross-sectional view showing the substrate processing apparatus according to the embodiment.

[0010] FIG. 4 is a horizontal cross-sectional view showing the substrate processing apparatus according to the embodiment.

[0011] FIG. 5 is a horizontal cross-sectional view showing the substrate processing apparatus according to the embodiment.

[0012] FIG. 6 is a cross-sectional view showing a pipe connection structure.

[0013] FIG. 7 is an enlarged view of area A in FIG. 6.

[0014] FIG. 8 is a cross-sectional view showing a pipe connection method.

[0015] FIG. 9 is a cross-sectional view showing the pipe connection method.

[0016] FIG. 10 is a cross-sectional view showing the pipe connection method.

DETAILED DESCRIPTION

[0017] Reference will now be made in detail to various embodiments, examples of which are illustrated in the accompanying drawings. In the following detailed description, numerous specific details are set forth in order to provide a thorough understanding of the present disclosure. However, it will be apparent to one of ordinary skill in the art that the present disclosure may be practiced without these specific details. In other instances, well-known methods, procedures, systems, and components have not been described in detail so as not to unnecessarily obscure aspects of the various embodiments.

[0018] Hereinafter, non-limiting exemplary embodiments of the present disclosure will be described with reference to the accompanying drawings. Throughout all the accompanying drawings, the same or corresponding members or components will be denoted by the same or corresponding reference numerals, and redundant descriptions thereof will be omitted.

Substrate Processing Apparatus

[0019] A substrate processing apparatus **1** according to an embodiment is described with reference to FIGS. **1** to **5**. FIG. **1** is a perspective view showing the substrate processing apparatus **1** according to the embodiment when the substrate processing apparatus **1** is viewed obliquely from above. FIG. **2** is a perspective view showing the substrate processing apparatus **1** according to the embodiment when the substrate processing apparatus **1** is viewed obliquely from below. FIG. **3** is a vertical cross-sectional view showing the substrate processing apparatus **1** according to the embodiment. FIG. **4** is a horizontal cross-sectional view showing the substrate processing apparatus **1** according to the embodiment, taken along line IV-IV in FIG. **3** in an arrow direction. FIG. **5** is a horizontal cross-sectional view showing the substrate processing apparatus **1** according to the embodiment, taken along line V-V in FIG. **3** in an arrow direction.

[0020] The substrate processing apparatus **1** is a batch-type apparatus that collectively performs various processes on a plurality of substrates. The various processes include, for example, a film formation process for forming films on the substrates by atomic layer deposition (ALD) or chemical vapor deposition (CVD). The various processes may include an etching process for removing the films formed on the substrates.

[0021] The substrate processing apparatus **1** includes a reaction tube **10**, a gas introducer **20**, a vacuum pipe **30**, an exhaust duct **40**, a housing **50**, a heating portion **60**, a depressurizer **70**, a pressurizer **80**, and an apparatus housing **90**. In FIGS. **1** and **2**, the housing **50**, the heating portion **60**, the depressurizer **70**, the pressurizer **80**, and the apparatus housing **90** are not illustrated. The reaction tube **10**, the gas introducer **20**, and the exhaust duct **40** are joined together by, for example, welding, and are integrally formed. The reaction tube **10**, the vacuum pipe **30**, and the exhaust duct **40** are formed of, for example, quartz.

[0022] The reaction tube **10** has a cylindrical shape with an open lower end and a ceiling. An introduction opening **10a** and an exhaust opening **10b** are provided on an outer wall of the reaction tube **10**.

[0023] The introduction opening **10a** penetrates the outer wall of the reaction tube **10**. The introduction opening **10a** is provided at positions at which gas introduction ducts **211** to **218**, which will be described later, are installed in a circumferential direction of the reaction tube **10**. A plurality of introduction openings **10a** is provided in a vertical direction from near an upper end to near a lower end of the reaction tube **10** at each position in the circumferential direction of the reaction tube **10**. In this case, it is easy to uniformly supply gas to a range from the upper end to the lower end of the reaction tube **10**.

[0024] The exhaust opening **10b** penetrates the outer wall of the reaction tube **10**. The exhaust opening **10b** is provided at a position different from the introduction opening **10a** in the circumferential direction of the reaction tube **10**. The exhaust opening **10b** is provided at a position at which the exhaust duct **40** is installed in the circumferential direction of the reaction tube **10**. The exhaust opening **10b** is a rectangular opening that extends in the vertical direction from near the upper end to near the lower end of the reaction tube **10**. In this case, it is easy to uniformly exhaust

gas over a range from the upper end to the lower end of the reaction tube **10**.

[0025] A lower end opening of the reaction tube **10** is hermetically sealed by a cover (not shown). The cover is formed of, for example, metal such as stainless steel. A substrate holder **11** (FIG. **3**) is accommodated inside the reaction tube **10**. The substrate holder **11** holds a plurality of substrates in a horizontal posture arranged in multiple stages in the vertical direction. The number of substrates is not limited, but is, for example, between **25** and **200**. The substrate holder **11** is formed of, for example, quartz.

[0026] The gas introducer **20** includes gas introduction ducts **211** to **218**, nozzles **221** to **228**, gas introduction pipes **231** to **238**, and opening/closing valves **241** to **248**. In FIG. **3**, the gas introduction pipes **231** to **235** and the opening/closing valves **241** to **245** are shown.

[0027] The gas introduction ducts **211** to **218** are provided in the circumferential direction of the reaction tube **10**. The gas introduction ducts **211** to **218** are provided to be spaced apart from each other in the circumferential direction of the reaction tube **10**. In this case, the adjacent gas introduction ducts **211** to **218** can have a reduced thermal effect on each other. This can suppress a temperature drop within the gas introduction ducts **211** to **218** and suppress the generation of particles. The gas introduction ducts **211** to **218** are radially provided to be spaced apart from each other in the circumferential direction of the reaction tube **10**. In this case, as indicated by arrows in FIG. **5**, gases such as a raw material gas, a reaction gas, an etching gas, and a purge gas can be supplied to the inside of the reaction tube **10** from a plurality of positions (in multiple directions) in the circumferential direction of the reaction tube **10**. Thus, an in-plane shape of a film formation or etching process can be easily adjusted. For example, a retention time of gas supplied to the surfaces of the substrates or a gas concentration distribution can be adjusted by adjusting a gas supply position or the amount of gas supplied. Therefore, it is easy to control the in-plane shape of the film formation or etching process compared to a unidirectional gas flow. The gas introduction ducts **211** to **218** are provided in this order counterclockwise from the exhaust opening **10b**.

[0028] The gas introduction ducts **211** to **218** are installed on the outer wall of the reaction tube **10**. In this case, a distance from the gas introduction ducts **211** to **218** to the substrates is shortened. Therefore, unnecessary thermal decomposition of the gas can be suppressed. In addition, when the gas introduction ducts **211** to **218** are installed on the outer wall of the reaction tube **10**, there is no need to dispose the nozzles **221** to **228** inside the reaction tube **10**. For this reason, a space between outer edges of the substrates and an inner wall of the reaction tube **10** can be narrowed, thereby reducing a gas flow in that space. As a result, the efficiency of gas supply between the vertically adjacent substrates is improved. Additionally, when the gas introduction ducts **211** to **218** are installed on the outer wall of the reaction tube **10**, there is no need to form a nozzle chamber for accommodating the nozzles **221** to **228** by protruding a side wall of the reaction tube **10** radially outward. The gas introduction ducts **211** to **218** are integrally formed with, for example, the reaction tube **10**. The gas introduction ducts **211** to **218** are formed of, for example, quartz.

[0029] The gas introduction ducts **211** to **218** have a tubular shape with closed lower ends and open upper ends. The upper ends of the gas introduction ducts **211** to **218** extend upward from an upper surface of the reaction tube **10** and penetrate the housing **50**. In this case, an upper space of the housing **50** can be used as a space for installing the gas introduction pipes **231** to **238** or the opening/closing valves **241** to **248**. Therefore, a pipe distance from the opening/closing valves **241** to **248** to the reaction tube **10** can be shortened. Further, the shape of the gas introduction pipes **231** to **238** can be simplified. Gas holes **211a** to **218a** (FIG. **5**) are provided at positions of the gas introduction ducts **211** to **218** facing the reaction tube **10**.

[0030] Each of the gas holes **211a** to **218a** has a rectangular shape extending in the vertical direction from near the upper end to near the lower end of the reaction tube **10**. Each of the gas holes **211a** to **218a** extends from above the uppermost introduction opening **10a** to below the lowermost introduction opening **10a**. Gas flowing inside the gas introduction ducts **211** to **218** is discharged into the reaction tube **10** from the gas holes **211a** to **218a**.

[0031] The gas introduction duct **211** is provided at an angle range less than 90° counterclockwise from the exhaust duct **40** in the circumferential direction of the reaction tube **10**. The gas introduction duct **212** is provided at an angle of 90° counterclockwise from the exhaust duct **40** in the circumferential direction of the reaction tube **10**. The gas introduction ducts **213** and **214** are provided at an angle range greater than 90° and less than 180° counterclockwise from the exhaust duct **40**. The gas introduction duct **215** is provided at an angle of 180° counterclockwise from the exhaust duct **40** in the circumferential direction of the reaction tube **10**. In other words, the gas introduction duct **215** is provided at a position opposite to the exhaust duct **40**. The gas introduction ducts **216** and **217** are provided at an angle range greater than 180° and less than 270° counterclockwise from the exhaust duct **40**. The gas introduction duct **218** is provided at an angle of 270° counterclockwise from the exhaust duct **40** in the circumferential direction of the reaction tube **10**. In other words, the gas introduction duct **218** is provided at a position opposite to the gas introduction duct **212**.

[0032] The nozzles **221** to **228** are detachably inserted into the interior of the gas introduction ducts **211** to **218**, respectively. In this case, the shape of the nozzles **221** to **228** can be changed according to the type of process, so that the optimal nozzles **221** to **228** can be used according to the type of process. An inner surface of each of the gas introduction ducts **211** to **218** has a shape that follows an outer surface of each of the nozzles **221** to **228**, and a gap is provided between the inner surface of each of the gas introduction ducts **211** to **218** and the outer surface of each of the nozzles **221** to **228**. This allows the volume of a space between the gas introduction ducts **211** to **218** and the nozzles **221** to **228** to be reduced. Therefore, the retention of gas in this space is suppressed, thereby improving the efficiency of gas supply to the substrates. In addition, since a surface area in contact with gas is reduced, the generation of particles in this space can be suppressed. Thus, cleaning this space is easy. In a cross-section perpendicular to a longitudinal direction of the nozzles **221** to **228**, the inner surface of the gas introduction ducts **211** to **218** is, for example, circular, and the outer surface of the nozzles **221** to **228** is, for example, circular. In the cross-section perpendicular to the longitudinal direction of the nozzles **221** to **228**, the inner surface of the gas introduction ducts **211** to **218** may be elliptical, and the outer surface of the nozzles **221** to **228** may also be elliptical.

[0033] The nozzles **221** to **228** are provided with gas discharge holes (not shown). The gas discharge holes are provided, for example, in portions of tube walls of the nozzles **221** to **228** inserted into the gas introduction ducts **211** to **218**. The upper ends of the nozzles **221** to **228** are connected to gas sources (not shown) via the gas introduction pipes **231** to **238**, respectively. Gases from the gas sources are introduced into the nozzles **221** to **228** from the upper ends of the nozzles **221** to **228** and discharged into the reaction tube **10** through the gas discharge holes, the gas holes **211a** to **218a**, and the introduction openings **10a**. The nozzles **221** to **228** are not illustrated in FIGS. 2 to 5.

[0034] The nozzles **221** to **228** may not be provided. In this case, the upper ends of the gas introduction ducts **211** to **218** are connected to the gas sources via the gas introduction pipes **231** to **238**, respectively. Gases from the gas sources are introduced into the gas introduction ducts **211** to **218** from the upper ends of the gas introduction ducts **211** to **218** and discharged into the reaction tube **10** through the gas holes **211a** to **218a**. For example, when gases that are easily thermally decomposed, such as hexachlorodisilane (HCD) gas and dichlorosilane (DCS) gas, are used, the nozzles **221** to **228** may not be necessary.

[0035] The gas introduction pipes **231** to **238** are provided in an upper space of the housing **50**. One end of each of the gas introduction pipes **231** to **238** is connected to each of the corresponding gas introduction ducts **211** to **218** or the corresponding nozzles **221** to **228**, and the other end thereof extends through the apparatus housing **90** to the exterior of the apparatus housing **90**. The gas introduction pipes **231** to **238** are provided with the opening/closing valves **241** to **248**. The gas introduction pipes **231** to **238** may also be provided with flow rate controllers such as mass flow

controllers.

[0036] The opening/closing valves **241** to **248** are provided in the upper space of the housing **50**. The opening/closing valves **241** to **248** are provided in the middle of the corresponding gas introduction pipes **231** to **238**. The opening/closing valves **241** to **248** are installed, for example, on an inner wall of the apparatus housing **90**. The opening/closing valves **241** to **248** are valves that switch a gas flow on and off.

[0037] The vacuum pipe **30** has a cylindrical shape with an open lower end and a ceiling. The vacuum pipe **30** is provided to be spaced apart from the reaction tube **10**. An opening **30a** (FIG. 5) is provided on an outer wall of the vacuum pipe **30** at the same position as the exhaust duct **40** in a circumferential direction of the vacuum pipe **30**. The opening **30a** is a rectangular opening extending in the vertical direction from near an upper end to near a lower end of the vacuum pipe **30**. A vertical length of the opening **30a** may be the same as a vertical length of the exhaust opening **10b**. The axis of the vacuum pipe **30** may be parallel to the axis of the reaction tube **10**. The lower end of the vacuum pipe **30** is connected to an exhaust device (not shown) such as a vacuum pump via a pipe (not shown). A cross-sectional area of a flow path of the vacuum pipe **30** may be equal to or greater than a cross-sectional area of a flow path of the exhaust duct **40**. In this case, exhaust flow velocity in the vertical direction becomes uniform, so that a uniform laminar flow in the vertical direction is formed. Thereby, inter-plane uniformity of the film formation or etching process is improved.

[0038] The exhaust duct **40** connects the reaction tube **10** and the vacuum pipe **30**. The exhaust duct **40** allows communication between the interior of the reaction tube **10** and the interior of the vacuum pipe **30**. One end of the exhaust duct **40** is connected to the outer wall of the reaction tube **10** to cover the exhaust opening **10b**, and the other end thereof is connected to the outer wall of the vacuum pipe **30** to cover the opening **30a**. In this case, compared to directly connecting the vacuum pipe **30** to the reaction tube **10** without providing the exhaust duct **40**, a length X (FIG. 4) occupied by the exhaust duct **40** in the circumferential direction of the reaction tube **10** can be shortened. As a result, an installable length of the gas introduction ducts **211** to **218** in the circumferential direction of the reaction tube **10** increases. Thus, the number of the gas introduction ducts **211** to **218** provided on the outer wall of the reaction tube **10** can be increased. The exhaust duct **40** may be divided into a plurality of parts in the vertical direction. In this case, a gas flow from the interior of the reaction tube **10** towards the vacuum pipe **30** is rectified. For this reason, the uniformity of the gas flow at different positions in the vertical direction inside the reaction tube **10** is improved.

[0039] The housing **50** accommodates the reaction tube **10**, the gas introducer **20**, the vacuum pipe **30**, the exhaust duct **40**, and the heating portion **60** therein. Since the housing **50** accommodates the heating portion **60** including heaters, the housing **50** is also called a heater shell. The housing **50** includes a bottom portion **51**, a ceiling portion **52**, and a side portion **53**.

[0040] The bottom portion **51** supports the reaction tube **10** and the vacuum pipe **30**. The ceiling portion **52** is provided above the upper surface of the reaction tube **10** and the upper surface of the vacuum pipe **30**. The ceiling portion **52** covers the upper surface of the reaction tube **10** and the upper surface of the vacuum pipe **30**. The side portion **53** is provided around the reaction tube **10**, the gas introducer **20**, the vacuum pipe **30**, and the exhaust duct **40**. The side portion **53** covers the reaction tube **10**, the gas introducer **20**, the vacuum pipe **30**, and the exhaust duct **40**. A lower end of the side portion **53** is connected to the bottom portion **51**, and an upper end thereof is connected to the ceiling portion **52**. The bottom portion **51**, the ceiling portion **52**, and the side portion **53** are configured, for example, separately. The bottom portion **51**, the ceiling portion **52**, and the side portion **53** may be integrally formed.

[0041] The heating portion **60** is provided inside the housing **50**. The heating portion **60** includes a first side heater **61**, a second side heater **62**, a third side heater **63**, a first ceiling heater **64**, a second ceiling heater **65**, and a lower heater **66**. The first side heater **61**, the second side heater **62**, the third side heater **63**, the first ceiling heaters **64**, the second ceiling heaters **65**, and the lower heater **66**

are, for example, carbon wire heaters. In this case, the substrates accommodated inside the reaction tube **10** can be rapidly heated and cooled.

[0042] A plurality of first side heaters **61** is provided around the reaction tube **10**. The plurality of first side heaters **61** is radially provided at intervals in the circumferential direction of the reaction tube **10**. Each of the first side heaters **61** is positioned at a different location from the exhaust duct **40** in the circumferential direction of the reaction tube **10**. Each of the first side heaters **61** may be divided into a plurality of parts in the vertical direction. In this case, temperature in the vertical direction can be independently adjusted by independently controlling the first side heaters **61** divided into the plurality of parts. As indicated by solid arrows in FIG. 4, the first side heaters **61** heat the substrates accommodated inside the reaction tube **10** from the exterior of the reaction tube **10** through thermal radiation.

[0043] The second side heater **62** is provided in a different position from the first side heater **61** in the circumferential direction of the reaction tube **10**. The second side heater **62** is provided at a different position from the plurality of gas introduction ducts **211** to **218** in the circumferential direction of the reaction tube **10**. The second side heater **62** is provided to include the same position as the exhaust duct **40** in the circumferential direction of the reaction tube **10**. Since the exhaust duct **40** is provided around the reaction tube **10** at the same position as the exhaust duct **40** in the circumferential direction of the reaction tube **10**, the second side heater **62** cannot be disposed around the reaction tube **10**. Therefore, the second side heater **62** is provided around the vacuum pipe **30**. In other words, the second side heater **62** is provided at a position to which a distance from a center C1 of the reaction tube **10** is farther than a distance from the center C1 to the first side heater **61**. For example, the second side heater **62**, when viewed in a plane, is provided to include a virtual half straight line L extending through a center C3 of the vacuum pipe **30** starting from the center C1 of the reaction tube **10**. The second side heater **62** may be divided into a plurality of parts in the vertical direction. In this case, temperature in the vertical direction can be independently adjusted by independently controlling the second side heater **62** divided into the plurality of parts. As indicated by a broken line arrow in FIG. 4, the second side heater **62** heats the vacuum pipe **30** and the exhaust duct **40** through thermal radiation and also heats the substrates accommodated inside the reaction tube **10**. Thus, the substrates accommodated inside the reaction tube **10** is heated from all directions around the reaction tube **10** by the first side heater **61** and the second side heater **62**. Therefore, the in-plane temperature uniformity of a substrate is improved.

[0044] A plurality of third side heaters **63** is provided around the vacuum pipe **30**. The plurality of third side heaters **63** is provided at intervals in the circumferential direction of the vacuum pipe **30**. Each third side heater **63** is positioned at a position different from the second side heater **62** in the circumferential direction of the vacuum pipe **30**. Each third side heater **63**, when viewed in a plane, is provided without including the virtual half straight line L. Each third side heater **63** may be divided into a plurality of parts in the vertical direction. In this case, temperature in the vertical direction can be independently adjusted by independently controlling the third side heater **63** divided into the plurality of parts. As indicated by dash-dotted line arrows in FIG. 4, the third side heaters **63** heat the vacuum pipe **30**.

[0045] The first ceiling heater **64** is provided between the upper surface of the reaction tube **10** and the ceiling portion **52** of the housing **50**. The first ceiling heater **64** heats the substrates accommodated inside the reaction tube **10** from above the reaction tube **10** through thermal radiation. The number of first ceiling heaters **64** may be one or may be two or more.

[0046] The second ceiling heater **65** is provided between the upper surface of the vacuum pipe **30** and the ceiling portion **52** of the housing **50**. The second ceiling heater **65** heats the vacuum pipe **30** from above the vacuum pipe **30** through thermal radiation. The number of second ceiling heaters **65** may be one or may be two or more.

[0047] A plurality of lower heaters **66** is provided around the lower portion of the reaction tube **10**. The plurality of the lower heaters **66** is provided radially at intervals in the circumferential

direction of the reaction tube **10**. The lower heaters **66** are positioned below the substrate holder **11**. The lower heaters **66** heat the lower portion of the reaction tube **10** through thermal radiation and suppress heat dissipation from the lower end opening of the reaction tube **10**.

[0048] The depressurizer **70** reduces pressure inside the housing **50**. The depressurizer **70** includes a pipe **71**, a safety valve **72**, an opening/closing valve **73**, and a vacuum pump **74**.

[0049] The pipe **71** is connected to a port **53a** provided in the side portion **53** of the housing **50**. The pipe **71** extends through the side portion **53** and the apparatus housing **90** to the exterior of the apparatus housing **90**. In the pipe **71**, the safety valve **72**, the opening/closing valve **73**, and the vacuum pump **74** are provided in order from the housing **50** side. The safety valve **72**, the opening/closing valve **73**, and the vacuum pump **74** are provided, for example, outside the apparatus housing **90**. The safety valve **72**, the opening/closing valve **73**, and the vacuum pump **74** may be provided outside the housing **50** but inside the apparatus housing **90**.

[0050] The safety valve **72** is changed to an open state from a closed state when pressure inside the housing **50** exceeds a set pressure, thereby maintaining the internal pressure of the housing **50** below the set pressure.

[0051] The opening/closing valve **73** is a valve that switches a gas flow on and off.

[0052] The vacuum pump **74** reduces pressure inside the housing **50** through the pipe **71**.

[0053] When the opening/closing valve **73** is changed to an open state, the interior of the housing **50** is depressurized by the vacuum pump **74**. When the interior of the housing **50** is depressurized, heat transfer by convection is suppressed. This suppresses heat transfer to the exterior of the housing **50**, so that a space above the housing **50** can be used to install the opening/closing valves **241** to **248**. In addition, since a thermal insulation material becomes unnecessary, a distance between the reaction tube **10** and the housing **50** can be shortened. Thus, a distance from the opening/closing valves **241** to **248** to the substrates can be shortened and thus gas controllability is improved. Furthermore, when the side heaters (the first side heater **61**, the second side heater **62**, and the third side heater **63**) are divided into a plurality of parts in the vertical direction, convection does not influence the interior of the housing **50**, so that heaters have little influence on one another in the vertical direction. Therefore, inter-plane (vertical direction) temperature controllability is improved. This makes it easy to selectively heat only the lower portion of the reaction tube **10**, selectively heat only the center of the reaction tube **10**, or selectively heat only the upper portion of the reaction tube **10**.

[0054] The pressurizer **80** restores the internal pressure of the depressurized housing **50** to atmospheric pressure. The pressurizer **80** includes a pipe **81**, a gas source **82**, a flow rate controller **83**, and an opening/closing valve **84**.

[0055] The pipe **81** is connected to a port **53b** provided in the side portion **53** of the housing **50**. In the pipe **81**, the gas source **82**, the flow rate controller **83**, and the opening/closing valve **84** are provided in order from an upstream side to a downstream side of a gas flow direction. The gas source **82**, the flow rate controller **83**, and the opening/closing valve **84** are provided, for example, outside the apparatus housing **90**. The gas source **82**, the flow rate controller **83**, and the opening/closing valve **84** may be provided outside the housing **50** but inside the apparatus housing **90**.

[0056] The gas source **82** is, for example, a supply source of an inert gas. The inert gas is, for example, nitrogen gas. The inert gas may be argon gas.

[0057] The flow rate controller **83** controls the flow rate of gas flowing through the pipe **81**. The flow rate controller **83** is, for example, a mass flow controller.

[0058] The opening/closing valve **84** is a valve that switches a gas flow on and off.

[0059] When the opening/closing valve **84** is changed to an open state, the flow rate of gas from the gas source **82** is controlled by the flow rate controller **83**, and the gas with the controlled flow rate is supplied to the interior of the housing **50**. Thus, the internal pressure of the depressurized housing **50** is restored to atmospheric pressure.

[0060] The apparatus housing **90** surrounds the housing **50**. The apparatus housing **90** covers the entirety of the housing **50**. The apparatus housing **90** supports the bottom portion **51** of the housing **50**.

[0061] As described above, according to the substrate processing apparatus **1** of the embodiment, the reaction tube **10**, the gas introducer **20**, the vacuum pipe **30**, and the exhaust duct **40** are accommodated inside the housing **50**, and the heating portion **60** is provided inside the housing **50**. This allows the substrates accommodated inside the reaction tube **10** to be heated from all directions around the reaction tube **10**. Thus, the in-plane temperature uniformity of a substrate can be improved. As a result, processing uniformity of the substrate is raised.

Pipe Connection Structure

[0062] A pipe connection structure in which the vacuum pipe **30** and an exhaust pipe **110** are connected with each other will now be described with reference to FIGS. **6** and **7**. FIG. **6** is a cross-sectional view showing the pipe connection structure. FIG. **7** is an enlarged view of area A in FIG. **6**.

[0063] The pipe connection structure is provided through the bottom portion **51** of the housing **50**. The bottom portion **51** has an opening **51h** at a position through which a lower end of the vacuum pipe **30** is inserted. The opening **51h** includes a first opening and a second opening. The first opening is provided at an upper portion of the opening **51h** and has a first opening diameter. The second opening is provided at a lower portion of the opening **51h** and has a second opening diameter larger than the first opening diameter. The bottom portion **51** includes a small-diameter portion **51a** forming the first opening and a large-diameter portion **51b** forming the second opening. The pipe connection structure includes the vacuum pipe **30**, the exhaust pipe **110**, and a fixing member **120**.

[0064] The vacuum pipe **30** is provided inside the housing **50**. The lower end of the vacuum pipe **30** is inserted through the opening **51h**. Since an opening diameter of the opening **51h** is larger than an outer diameter of the vacuum pipe **30**, a gap **G1** is formed between the opening **51h** and the vacuum pipe **30**. In this case, since the bottom portion **51** and the vacuum pipe **30** are separated by the gap **G1**, damage to the vacuum pipe **30** can be prevented.

[0065] The exhaust pipe **110** is formed of, for example, a nickel alloy. The exhaust pipe **110** is provided below the bottom portion **51**. The exhaust pipe **110** is provided directly below the vacuum pipe **30**. The exhaust pipe **110** is provided coaxially or approximately coaxially with the vacuum pipe **30**. The interior of the exhaust pipe **110** is in communication with the interior of the vacuum pipe **30**. The exhaust pipe **110** includes a pipe portion **111** and a pipe flange portion **112**. The pipe flange portion **112** is provided at an upper portion of the pipe portion **111**. The pipe flange portion **112** protrudes radially outward from an outer side surface of the pipe portion **111**. The pipe flange portion **112** has, for example, an annular plate shape. An end of the pipe portion **111** that is not provided with the pipe flange portion **112** is connected to a vacuum pump (not shown). The vacuum pump exhausts gas introduced into the reaction tube **10** through the exhaust pipe **110**, the vacuum pipe **30**, and the exhaust duct **40**.

[0066] The fixing member **120** fixes the exhaust pipe **110** to the bottom portion **51**. The fixing member **120** includes a first flange **121**, a second flange **122**, and a clamp **123**.

[0067] The first flange **121** is formed of, for example, stainless steel. The first flange **121** includes a pipe portion **121a** and a flange portion **121b**. The first flange **121** is fixed to the bottom portion **51** by a screw **V1** inserted through the flange portion **121b** in a state in which the pipe portion **121a** is fitted into an inner side of the large-diameter portion **51b**.

[0068] An outer diameter of the pipe portion **121a** is larger than an opening diameter of the small-diameter portion **51a**. An upper surface of the pipe portion **121a** contacts a lower surface of the small-diameter portion **51a**. An O-ring **R1**, which is a sealing member, is provided at a contact portion between the upper surface of the pipe portion **121a** and the lower surface of the small-diameter portion **51a**. The O-ring **R1** contacts the upper surface of the pipe portion **121a** and the

lower surface of the small-diameter portion **51a** to seal a gap between the upper surface of the pipe portion **121a** and the lower surface of the small-diameter portion **51a**. The O-ring **R1** hermetically separates the interior of the housing **50** and the exterior of the housing **50**.

[0069] The outer diameter of the pipe portion **121a** is smaller than the opening diameter of the large-diameter portion **51b**. A gap **G2** is provided between an outer side surface of the pipe portion **121a** and an inner side surface of the large-diameter portion **51b**. In this case, even if a central axis of the vacuum pipe **30** is misaligned with a central axis of the opening **51h** due to tolerances or manufacturing errors of quartz, misalignment therebetween can be absorbed by the gap **G2**.

[0070] The pipe portion **121a** has an inner inclined surface **121c**, an inner diameter of which widens from top to bottom. The pipe portion **121a** has an inner side surface **121d**, the inner diameter of which is constant from top to bottom. The inner side surface **121d** is positioned below the inner inclined surface **121c**. A boundary between the inner inclined surface **121c** and the inner side surface **121d** is located above the lower end of the vacuum pipe **30**.

[0071] An inner diameter of the pipe portion **121a** at the upper end thereof is larger than an outer diameter of the vacuum pipe **30**. A gap **G3** is provided between an inner side surface of the pipe portion **121a** at the upper end thereof and an outer side surface of the vacuum pipe **30**. Since the first flange **121** and the vacuum pipe **30** are separated by the gap **G3**, damage to the vacuum pipe **30** can be prevented.

[0072] The flange portion **121b** is provided at a lower portion of the pipe portion **121a**. The flange portion **121b** protrudes radially outward from the outer side surface of the pipe portion **121a**. The flange portion **121b** has, for example, an annular plate shape. The flange portion **121b** is provided with an insertion through-hole **121h** through which the screw **V1** is inserted. A plurality of insertion through-holes **121h** is provided at intervals in the circumferential direction of the flange portion **121b**. The first flange **121** is fixed to the bottom portion **51** by the screw **V1** inserted through the insertion through-hole **121h**.

[0073] The first flange **121** is provided with a refrigerant flow path **121f** for circulating a

[0074] refrigerant. By circulating the refrigerant through the refrigerant flow path **121f**, the O-ring **R1** can be cooled via the first flange **121**. Thereby, even if the bottom portion **51** is heated due to thermal radiation, the O-ring **R1** contacting the bottom portion **51** is prevented from reaching a high temperature.

[0075] The second flange **122** is formed of, for example, a nickel alloy. The second flange **122** includes a pipe portion **122a** and a flange portion **122b**. The second flange **122** is fixed to the first flange **121** by a screw **V2** inserted through the flange portion **122b** in a state in which the pipe portion **122a** is fitted into an inner side of the pipe portion **121a**.

[0076] An upper end of the pipe portion **122a** is positioned above the lower end of the vacuum pipe **30**. The upper end of the pipe portion **122a** is positioned approximately at the same height as the boundary between the inner inclined surface **121c** and the inner side surface **121d**.

[0077] An outer diameter of the pipe portion **122a** is slightly smaller than an inner diameter of the inner side surface **121d** of the pipe portion **121a**. An outer side surface of the pipe portion **122a** has a curved shape along the inner side surface **121d** of the pipe portion **121a**. In this case, the second flange **122** can be easily fitted into the first flange **121**.

[0078] An inner diameter of the pipe portion **122a** is constant in the vertical direction. The inner diameter of the pipe portion **122a** is larger than the outer diameter of the vacuum pipe **30**. A gap **G4** is provided between an inner side surface of the pipe portion **122a** and the outer side surface of the vacuum pipe **30**. In this case, since the pipe portion **122a** and the vacuum pipe **30** are separated by the gap **G4**, damage to the vacuum pipe **30** can be prevented.

[0079] An O-ring **R2**, which is a sealing member, is provided in an area surrounded by the outer side surface of the vacuum pipe **30**, the inner inclined surface **121c** of the first flange **121**, and the upper end of the second flange **122**. The O-ring **R2** contacts the outer side surface of the vacuum pipe **30**, the inner inclined surface **121c** of the first flange **121**, and the upper end of the second

flange **122**, thereby sealing the gaps **G3** and **G4**. The O-ring **R2** hermetically separates the interior of the vacuum pipe **30**, the interior of the housing **50**, and the exterior of the housing **50** from each other. The inner diameter of the pipe portion **122a** may be dimensioned such that the O-ring **R2** does not protrude from the gap **G4** under vacuuming.

[0080] The flange portion **122b** is provided at a lower portion of the pipe portion **122a**. The flange portion **122b** protrudes radially outward from the outer side surface of the pipe portion **122a**. The flange portion **122b** has, for example, an annular plate shape. The outer diameter of the flange portion **122b** is the same as, for example, the outer diameter of the flange portion **121b**. An upper surface of the flange portion **122b** contacts a lower surface of the flange portion **121b**. The flange portion **122b** is provided with an insertion through-hole **122h** through which the screw **V2** is inserted. A plurality of insertion through-holes **122h** is provided at intervals in the circumferential direction of the flange portion **122b**. The second flange **122** is fixed to the first flange **121** by the screws **V2** inserted through the insertion through-holes **122h**.

[0081] The clamp **123** is formed of, for example, stainless steel. The clamp **123** includes a pipe portion **123a** and a pressing portion **123b**. The clamp **123** is fixed to the flange portion **122b** by a screw **V3** by inserting the pipe flange portion **112** between the clamp **123** and the lower surface of the flange portion **122b** by the pressing portion **123b**.

[0082] An outer diameter of the pipe portion **123a** is the same as, for example, an outer diameter of the flange portion **122b**. An inner diameter of the pipe portion **123a** is larger than an outer diameter of the pipe flange portion **112**. A gap **G5** is provided between an inner side surface of the pipe portion **123a** and an outer side surface of the pipe flange portion **112**. In this case, even if the central axis of the vacuum pipe **30** is misaligned with the central axis of the exhaust pipe **110**, misalignment therebetween can be absorbed by the gap **G5**. Therefore, the exhaust pipe **110** can be easily installed while ensuring airtightness. The pipe portion **123a** is provided with an insertion through-hole **123h** through which the screw **V3** is inserted. A plurality of insertion through-holes **123h** is provided at intervals in the circumferential direction of the pipe portion **123a**. The clamp **123** is fixed to the second flange **122** by the screw **V3** inserted through the insertion through-hole **123h**.

[0083] The pressing portion **123b** is provided at the lower portion of the pipe portion **123a**. The pressing portion **123b** protrudes radially inward from the inner side surface of the pipe portion **123a**. The pressing portion **123b** has, for example, an annular plate shape. An inner diameter of the pressing portion **123b** is larger than an outer shape of the pipe portion **111** of the exhaust pipe **110**.

[0084] An O-ring **R3**, which is a sealing member, is provided at a contact portion at which the lower surface of the flange portion **122b** contacts the upper surface of the pipe flange portion **112**. The O-ring **R3** contacts the lower surface of the flange portion **122b** and the upper surface of the pipe flange portion **112**, thereby sealing a gap between the lower surface of the flange portion **122b** and the upper surface of the pipe flange portion **112**. The O-ring **R3** hermetically separates the interior of the vacuum pipe **30** from the exterior of the housing **50**.

Pipe Connection Method

[0085] A pipe connection method for connecting the vacuum pipe **30** and the exhaust pipe **110** will now be described with reference to FIGS. **8** to **10**. FIGS. **8** to **10** are cross-sectional views showing the pipe connection method.

[0086] First, the lower end of the reaction tube **10** is fixed to the bottom portion **51** (FIG. **3**). As a result, as shown in FIG. **8**, the lower end of the vacuum pipe **30**, which is integrally formed with the reaction tube **10**, is inserted into the opening **51h**. In this case, since the opening diameter of the opening **51h** is larger than the outer diameter of the vacuum pipe **30**, the gap **G1** is formed between the opening **51h** and the vacuum pipe **30**. In this case, since the bottom portion **51** and the vacuum pipe **30** are separated by the gap **G1**, damage to the vacuum pipe **30** can be prevented.

[0087] Next, as shown in FIG. **8**, the O-ring **R1** is installed on the lower surface of the small-diameter portion **51a**, the first flange **121** is installed at the lower end of the vacuum pipe **30**, and

the first flange **121** is fixed to the bottom portion **51** by the screw **V1**. In this case, since the outer diameter of the pipe portion **121a** is smaller than the opening diameter of the large-diameter portion **51b**, the gap **G2** is formed between the outer side surface of the pipe portion **121a** and the inner side surface of the large-diameter portion **51b**. In this case, even if the central axis of the vacuum pipe **30** is misaligned with the central axis of the opening **51h** due to tolerances or manufacturing errors of quartz, misalignment therebetween can be absorbed by the gap **G2**.

[0088] Next, as shown in FIG. **9**, the O-ring **R2** is installed between the outer side surface of the vacuum pipe **30** and an inner side surface of the first flange **121**. Then, the second flange **122** is fixed to the first flange **121** by the screw **V2** by fitting the pipe portion **122a** of the second flange **122** into an inner side of the first flange **121**. Since the outer side surface of the pipe portion **122a** has a curved shape along the inner side surface **121d** of the pipe portion **121a**, the second flange **122** can be easily fitted into the first flange **121**.

[0089] Next, as shown in FIG. **10**, the pipe flange portion **112** of the exhaust pipe **110** is brought into contact with the lower surface of the second flange **122** through the O-ring **R3**. Then, the pipe flange portion **112** is inserted between and the clamp **123** and the lower surface of the flange portion **122b** by the pressing portion **123b** of the clamp **123**, and the clamp **123** is fixed to the second flange **122** by the screw **V3**. In this case, since the inner diameter of the pipe portion **123a** is larger than the outer diameter of the pipe flange portion **112**, the gap **G5** is formed between the inner side surface of the pipe portion **123a** and the outer side surface of the pipe flange portion **112**. In this case, even if the central axis of the vacuum pipe **30** is misaligned with the central axis of the exhaust pipe **110**, misalignment therebetween can be absorbed by the gap **G5**. Therefore, the exhaust pipe **110** can be easily installed while ensuring airtightness.

[0090] It should be noted that the embodiments disclosed herein are exemplary in all aspects and are not restrictive. The above-described embodiments may be omitted, replaced, or modified in various forms without departing from the scope and spirit of the appended claims.

[0091] According to the present disclosure in some embodiments, it is possible to easily install the reaction tube and the vacuum pipe that are integrally formed.

[0092] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosures. Indeed, the embodiments described herein may be embodied in a variety of other forms. Furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the disclosures. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosures.

Claims

1. A substrate processing apparatus, comprising: a reaction tube; a vacuum pipe integrally formed with the reaction tube; a housing configured to accommodate the reaction tube and the vacuum pipe therein; an exhaust pipe provided below the housing; and a fixing member configured to fix the exhaust pipe to the housing, wherein the housing has a bottom portion configured to support the reaction tube, the bottom portion has an opening with an opening diameter larger than an outer diameter of the vacuum pipe, the vacuum pipe has a lower end inserted through the opening, and the exhaust pipe is fixed to the housing in a state in which an interior thereof communicates with an interior of the vacuum pipe.
2. The substrate processing apparatus of claim 1, wherein a pipe axis of the vacuum pipe is parallel to a pipe axis of the reaction tube.
3. The substrate processing apparatus of claim 1, wherein an upper portion of the exhaust pipe is provided with a flange portion protruding radially outward, and wherein the fixing member includes: a first flange disposed around a lower end of the vacuum pipe with a first gap and fixed to

the bottom portion; a second flange disposed on an inner side of the first flange and fixed to the first flange; and a clamp fixed to the second flange by inserting the flange portion between the clamp and the second flange.

4. The substrate processing apparatus of claim 3, wherein the fixing member has a first sealing member that contacts the bottom portion and the first flange, and hermetically separates an interior of the housing and an exterior of the housing from each other.

5. The substrate processing apparatus of claim 3, wherein the fixing member has a second sealing member that contacts an outer side surface of the vacuum pipe, an inner side surface of the first flange, and an upper end of the second flange, and hermetically separates the interior of the vacuum pipe, an interior of the housing, and an exterior of the housing from each other.

6. The substrate processing apparatus of claim 3, wherein the fixing member has a third sealing member that contacts the second flange and the flange portion, and hermetically separates an exterior of the housing and the interior of the vacuum pipe from each other.

7. The substrate processing apparatus of claim 3, wherein a second gap is provided between an outer side surface of the first flange and an inner surface of the opening.

8. The substrate processing apparatus of claim 3, wherein a third gap is provided between an outer side surface of the flange portion and an inner side surface of the clamp.

9. The substrate processing apparatus of claim 3, wherein a refrigerant flow path is provided inside the first flange.

10. The substrate processing apparatus of claim 1, wherein the reaction tube and the vacuum pipe are formed of quartz.
